

Pranav Pappu

pranavpappu@u.nus.edu | +65 9754 0582 | LinkedIn | GitHub | pranavpappu.dev | X

Availability: Jan–Dec 2027 for internships in the San Francisco Bay Area

Singapore citizen; eligible for 12-month J-1 visa sponsored by NUS; program hosted by Stanford University.

AI Engineer & Founder specializing in high-performance agentic systems (MCP). Reduced delivery timelines from weeks to days (Tweakler) and documentation overhead by 78% (AssessMate).

Education

National University of Singapore

2025 – 2029

B.Comp. (Hons) in Computer Science • **Teaching Assistant, CS1101S Programming Methodology**

AIE Scholarship – Selected as 1 of 20 recipients

Technical Skills

Languages/Runtime: Python, Zig, C++, Rust, TypeScript/JavaScript, Node.js

AI & Data: MCP (stdio/HTTP), PyTorch, Transformers, OpenAI API, pgvector, FastAPI, Pydantic

Systems & Infrastructure: Linux, Docker, GitHub Actions CI/CD, Fly.io, PostgreSQL, pytest, Playwright

Experience

AI Engineer Intern, Ebb & Flow

Sep 2026 – Present

- Building AI systems for restaurant HR workflows, including hiring, staff scheduling, and rostering.

Founder, Tweakler

2026 – Present

- Founded Tweakler, a reference-led site generator: transforms one-line briefs into editable, brand-specific sites by deriving audience values and surfacing diverse design references.
- Reduced website delivery timelines from 3 - 4 weeks to under 7 days for studios like Brightsea Studio by building a reference-led generation system with an MCP workflow.

Research Assistant, National University of Singapore

Jan 2026 – Jun 2026

- Systematized the lab's evaluation of 3DGS methods by building a dynamic benchmark with 11 YAML pipelines and a 6-metric evaluator, enabling consistent performance tracking across 6 method repositories.

CCSGP Fellow (AssessMate), NUS Centre for Computing for Social Good

2026 – Present

- Reduced case-note documentation time from 90 to 20 minutes per child (a 78% reduction) by building AssessMate, an AI copilot that processes PDPA-sensitive data for early childhood intervention assessors.

Projects

EvoServe – Agent Evaluation & Benchmark Construction

2026 – Present

- Built a deterministic offline verifier in Zig for coding-agent runs, validating captured evidence and producing auditable quality and regression reports.
- Designing an audience-driven benchmark pipeline that uses agents to generate task families and reproducible environments, guided by quality-diversity search.
- Defined admission criteria using independently grounded answers, adversarial grader validation, and held-out evaluation to measure harness improvements.

GPT-2 (124M) Recreation from Scratch

2024

- Recreated GPT-2 124M from scratch (tokenizer, transformer blocks, optimizer, mixed precision, and FineWeb Edu evaluation); trained on 10B tokens.

WikiBase – Source-Grounded Knowledge Platform

2026

- Built WikiBase, a student knowledge platform that turns PDFs and notes into cited Markdown wikis, grounded Q&A, and flashcards; architected its FastAPI/PostgreSQL/pgvector backend, idempotent 5-stage pipeline, Docker/Fly.io release infrastructure, and GitHub Actions CI/CD plus a fail-closed 487-test harness covering migrations, security scans, and smoke checks.

Competitions

Replit Agent Hackathon

2026

- 1st place: Built Remember, an AR memory assistant with 4 AI workflows for face recognition, speech transcription, LLM memory extraction, and voice Q&A over 128-float face embeddings to a persistent personal knowledge graph for context-aware recall across repeated encounters.

Google DeepMind × Cactus (YC S25) Global Hackathon

2026

- 2nd place: Built SecureClaw, a privacy router for 12 PII types with 3 routing modes, on-device/cloud inference, and a 30-case benchmark.